

# Multi-Agent Reinforcement Learning

Why MARL is not just bigger RL

- Other agents are adaptive parts of the environment
- Learning changes everyone else's data distribution
- Coordination, credit assignment, and exploration become coupled

**Single learner  
fixed world**

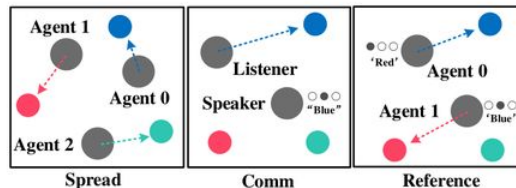
**Many learners  
moving world**

# Popular MARL Environments

Benchmarks stress different MARL failure modes: partial observability, coordination, competition, scale, and credit assignment.



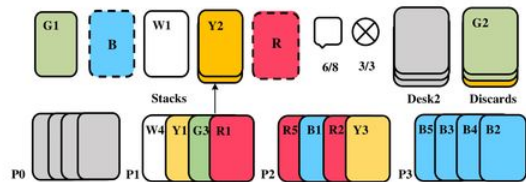
**SMAC / StarCraft II**



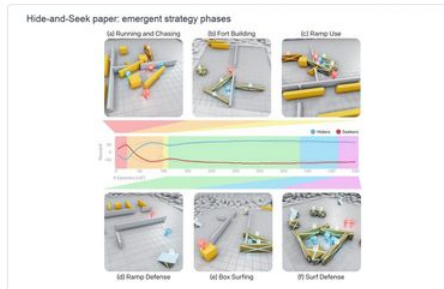
**Multi-Agent Particle Env.**



**Google Research Football**



**Hanabi**



**OpenAI Hide-and-Seek**



**Dota 2 / OpenAI Five**

# Single-Agent RL Assumes a Stationary MDP

- The policy changes during training
- The transition kernel  $P$  is fixed
- The reward function  $r$  is fixed
- PPO-style updates rely on this stationary data-generating process

$$\begin{aligned}\mathcal{M} &= (\mathcal{S}, \mathcal{A}, P, r, \gamma) \\ s_{t+1} &\sim P(\cdot \mid s_t, a_t) \\ J(\pi) &= \mathbb{E}_{\pi, P} \left[ \sum_{t=0}^T \gamma^t r(s_t, a_t) \right]\end{aligned}$$

*Key assumption:  $P$  and  $r$  do not depend on the training iteration  $k$ .*

# The Induced Environment for One Agent

- Agent  $i$  chooses only  $a_i$
- Other agents contribute  $a_{-i}$
- From agent  $i$ 's view, their policies are part of the transition model
- If those policies move, the induced environment moves

$$\mathbf{a}_t = (a_t^1, \dots, a_t^N)$$
$$P_i^{\pi_{-i}}(s' \mid s, a_i) = \sum_{a_{-i}} \pi_{-i}(a_{-i} \mid o_{-i}) P(s' \mid s, a_i, a_{-i})$$

*The physical simulator  $P$  can be fixed while the agent-specific effective transition changes.*

# Non-Stationarity in Independent Learning

- Each learner estimates returns from yesterday's teammates or opponents
- Another policy update can invalidate that estimate
- Replay, advantages, and evaluation can become stale faster than in single-agent RL

$$P_{i,k}(s' \mid s, a_i) = P_i^{\pi_{-i,k}}(s' \mid s, a_i)$$
$$\pi_{-i,k+1} \neq \pi_{-i,k} \Rightarrow P_{i,k+1} \neq P_{i,k}$$

*This is the formal version of the VMAS all-agents-learning problem.*

# Problem Structure vs Algorithm Structure

- Problem: what the environment exposes
- Algorithm: what the learner chooses to use
- Centralized training is not centralized control
- Execution constraints matter

local observation:  $o_i$    vs.   global state:  $s$

individual action:  $a_i$    vs.   joint action:  $\mathbf{a} = (a_1, \dots, a_N)$

individual reward:  $r_i$    vs.   team reward:  $r_{\text{team}}$

# Centralized Training, Decentralized Execution

- Actors remain deployable from local observations
- Critic can use privileged simulator information
- CTDE reduces variance without requiring centralized execution

decentralized actor:  $a_i \sim \pi_{\theta_i}(\cdot \mid o_i)$

centralized critic:  $V_\phi(s)$  or  $Q_\phi(s, a_1, \dots, a_N)$

execution constraint:  $\pi_{\theta_i} : o_i \mapsto \Delta(A_i)$

# Cooperative MARL as a Dec-POMDP

- Cooperative and partially observable
- Each agent acts from local observation  $o_i$
- Agents optimize one shared team return
- Shared reward still leaves coordination and credit assignment

hidden state  $s$

local views:

$o_1, o_2, \dots, o_N$

$$\mathcal{G}_{\text{Dec-POMDP}} = \langle S, \{A_i\}_{i=1}^N, P, \{O_i\}_{i=1}^N, Z, r, \gamma \rangle$$

$$o_i \sim Z(\cdot \mid s, i)$$

$$r_t = r(s_t, \mathbf{a}_t)$$

$$J(\pi) = \mathbb{E}_{\pi} \left[ \sum_{t=0}^T \gamma^t r_t \right]$$



# Markov Games and Individual Rewards

- Generalizes MDPs to multiple agents
- Each agent can have its own reward  $r_i$
- Competitive and mixed settings connect MARL to game theory
- Evaluation must consider opponents, equilibria, and exploitability

Reward structure

Cooperative

Zero-sum

Mixed

Condition

$r_1 = \dots = r_N$

$r_1 = -r_2$

rewards partly aligned

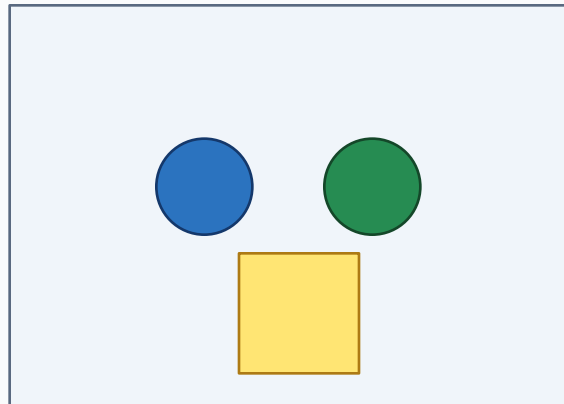
$$\mathcal{G}_{\text{Markov}} = \langle S, \{A_i\}_{i=1}^N, P, \{r_i\}_{i=1}^N, \gamma \rangle$$

$$J_i(\boldsymbol{\pi}) = \mathbb{E}_{\boldsymbol{\pi}} \left[ \sum_{t=0}^T \gamma^t r_i(s_t, \mathbf{a}_t) \right]$$

cooperative:  $r_1 = \dots = r_N$ ,      zero-sum:  $r_1 = -r_2$

# Shared Reward Does Not Solve Credit Assignment

- The team objective is aligned
- The learning signal is not agent-specific
- One reward must explain many local actions
- Agents can duplicate work or block each other



*Both agents affect one team reward*

$$r_i(t) = r_{\text{team}}(t) \quad \forall i$$
$$\nabla_{\theta_i} J \approx \mathbb{E} \left[ \nabla_{\theta_i} \log \pi_{\theta_i}(a_i \mid o_i) A^{\text{team}} \right]$$

# Counterfactual and Centralized Critics

- A centralized critic can see the joint situation
- Counterfactual baselines vary one agent's action
- The baseline asks: what if agent  $i$  had acted differently?
- Goal: lower-variance, more targeted learning signal

$$Q_i(s, a_1, \dots, a_N)$$

*hold  $a_{-i}$  fixed  
average over possible  $a_i$ '*

$$Q_i = Q_i(s, a_1, \dots, a_N)$$

$$A_i(s, \mathbf{a}) = Q_i(s, \mathbf{a}) - b_i(s, a_{-i})$$

$$b_i(s, a_{-i}) = \sum_{a'_i} \pi_i(a'_i \mid o_i) Q_i(s, (a'_i, a_{-i}))$$

# Parameter Sharing

- Natural when agents are homogeneous or exchangeable
- Improves sample efficiency by pooling experience
- Agent IDs or role features can break symmetry
- Separate policies are better for genuinely different roles

**one policy**  
 $\pi_{\theta}$

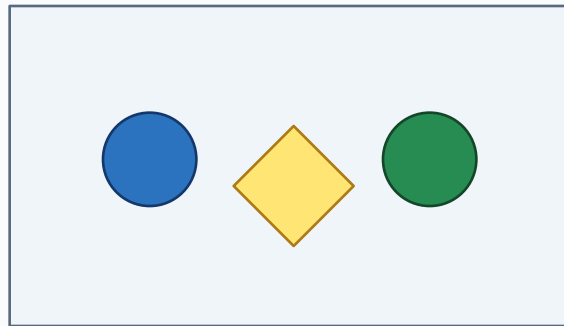
**N policies**  
 $\pi_{\theta_i}$

*Tradeoff:*  
*symmetry + data efficiency*  
*vs. role specialization*

shared policy:  $a_i \sim \pi_{\theta}(\cdot \mid o_i, \text{id}_i)$   
separate policies:  $a_i \sim \pi_{\theta_i}(\cdot \mid o_i)$

# Exploration Is Coupled Across Agents

- Useful behavior may require several agents to deviate together
- One agent exploring alone can look bad
- Entropy broadens one policy's action coverage
- Coordination may need correlated exploration, curriculum, or diversity



*Some outcomes need coordinated deviations*

$$\mathcal{H}(\pi_i(\cdot \mid o_i)) = - \sum_a \pi_i(a \mid o_i) \log \pi_i(a \mid o_i)$$
$$L_i = L_i^{PPO} + \alpha \mathbb{E}_t[\mathcal{H}(\pi_i(\cdot \mid o_i^t))]$$

# Algorithm Map

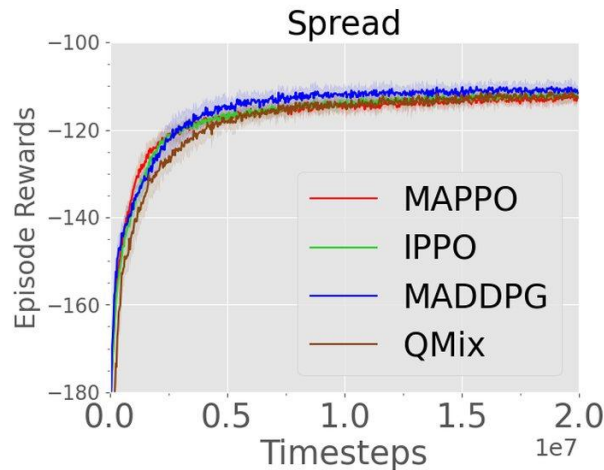
*Most MARL algorithms modify familiar RL around information, rewards, critics, and populations.*

| Family                  | Execution / actor         | Training / critic      | Main idea             |
|-------------------------|---------------------------|------------------------|-----------------------|
| IPPO                    | local PPO actors          | local value estimates  | simplest PPO baseline |
| MAPPO                   | local PPO actors          | centralized critic     | CTDE for cooperative  |
| PPO                     |                           |                        |                       |
| MADDPG / MASAC          | local continuous actors   | centralized Q critics  | CTDE actor-critic     |
| VDN / QMIX              | decentralized greedy acts | centralized team value | value decomposition   |
| Self-play / populations | policy pool sampling      | opponent/team pools    | robustness and        |
| curriculum              |                           |                        |                       |
| MADPO-style diversity   | local policies            | auxiliary divergence   | exploration           |
| heterogeneity           |                           |                        |                       |

# IPPO: PPO Per Agent

- Run PPO independently for each agent
- Easy to implement and often strong
- Each learner treats other learners as environment dynamics
- Main weakness: non-stationarity

MAPPO paper: IPPO vs MAPPO on MPE Spread



$$\rho_{i,t}(\theta_i) = \frac{\pi_{\theta_i}(a_{i,t} \mid o_{i,t})}{\pi_{\theta_i^{old}}(a_{i,t} \mid o_{i,t})}$$

$$L_i^{CLIP} = \mathbb{E}_t [\min(\rho_{i,t} A_{i,t}, \text{clip}(\rho_{i,t}, 1 - \epsilon, 1 + \epsilon) A_{i,t})]$$

# MAPPO: PPO with Centralized Value Learning

- Actors execute from local observations
- Critic trains with centralized information
- Better value estimates reduce variance
- This is the PPO version of the CTDE move

MAPPO paper: local, global, and overlapping critic inputs

|      |             |             |            |             |
|------|-------------|-------------|------------|-------------|
| IND: | Local Only  | Overlapping |            |             |
| EP:  | Global Only | Overlapping |            |             |
| AS:  | Global Only | Overlapping | Local Only | Overlapping |
| FP:  | Global Only | Overlapping | Local Only |             |

- Agent ID
- Available Actions
- Relative Distances
- Own information
- Visible ally/enemy info
- Out of sight ally/enemy info
- “Global” distances

$\pi_{\theta_i}(a_i | o_i)$  decentralized actor

$V_{\phi}(s)$  or  $V_{\phi}(o_1, \dots, o_N)$  centralized critic

$$A_t = R_t - V_{\phi}(s_t)$$

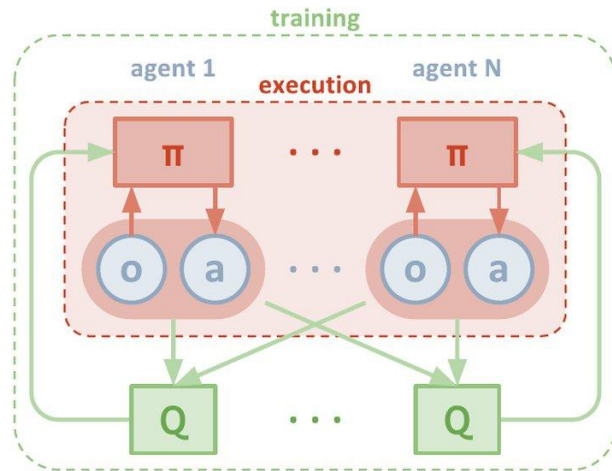


# MADDPG and Centralized Critics for Mixed Games

- Deterministic local actors
- Critics condition on joint observations/actions
- Designed for mixed cooperative-competitive settings
- MPE was the compact benchmark family

$$a_i = \mu_{\theta_i}(o_i)$$
$$Q_i^\mu = Q_i^\mu(x, a_1, \dots, a_N)$$
$$\nabla_{\theta_i} J_i \approx \mathbb{E} [\nabla_{\theta_i} \mu_i(o_i) \nabla_{a_i} Q_i^\mu(x, a_1, \dots, a_N)]$$

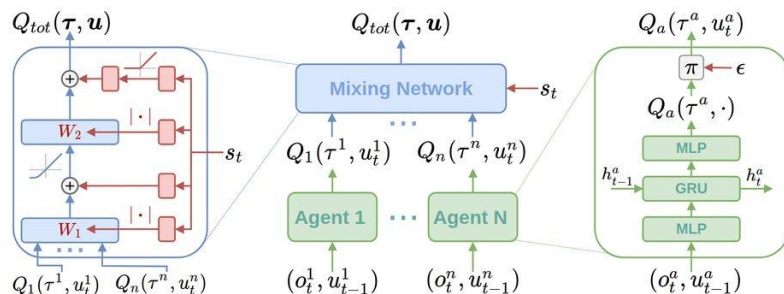
MADDPG paper: decentralized actors, centralized critics



# VDN and QMIX: Decomposing Team Value

- Cooperative discrete-action tasks
- Learn a centralized team value
- Preserve decentralized greedy action selection
- QMIX relaxes VDN with a monotonic mixing network

QMIX paper: per-agent utilities mixed into  $Q_{tot}$



$$\text{VDN: } Q_{tot}(\boldsymbol{\tau}, \mathbf{a}) = \sum_i Q_i(\tau_i, a_i)$$

$$\text{QMIX: } Q_{tot} = f_{\phi}(Q_1, \dots, Q_N, s), \quad \frac{\partial Q_{tot}}{\partial Q_i} \geq 0$$

$$\arg \max_{\mathbf{a}} Q_{tot} = (\arg \max_{a_1} Q_1, \dots, \arg \max_{a_N} Q_N)$$

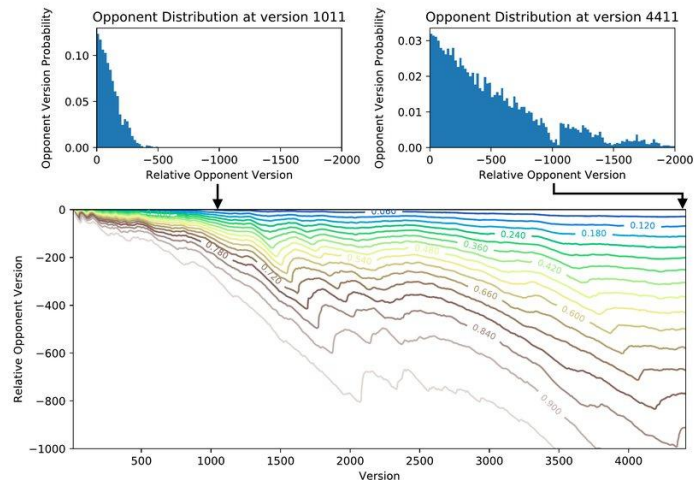
# Self-Play and Populations

- Opponents become the curriculum
- Train against current or past policies
- Populations reduce overfitting to one opponent
- Risk: non-transitivity and hidden exploitability

$$J(\pi_k) = \mathbb{E}_{\pi_k, \pi_{opp} \sim \mathcal{P}_k} \left[ \sum_{t=0}^T \gamma^t r_t \right]$$

$$\mathcal{P}_{k+1} = \mathcal{P}_k \cup \{\pi_k\}$$

OpenAI Five paper: opponent sampling over self-play



# OpenAI Five: Large-Scale Self-Play

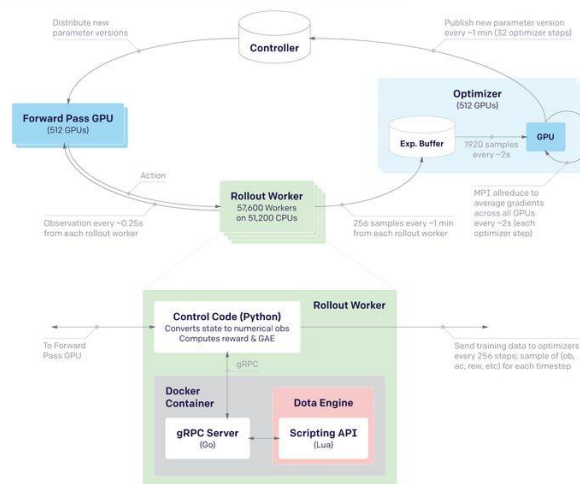
- Domain: Dota 2, 5v5 team competition
- Large-scale self-play over long horizons
- Team coordination from shared objectives
- Reward shaping, curriculum, and distributed systems were part of the method

$$\pi_{opp} \sim \mathcal{P}_k$$

$$\mathcal{D}_k = \{\tau_m \sim (\pi_k, \pi_{opp})\}_{m=1}^M$$

$$\pi_{k+1} \leftarrow \text{Update}(\pi_k, \mathcal{D}_k)$$

OpenAI Five paper: distributed training architecture

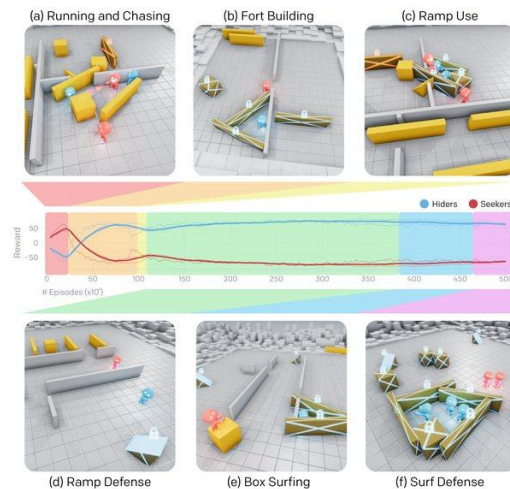


# OpenAI Hide-and-Seek: Autocurricula

- Competition generates an automatic curriculum
- Hiders improve, then seekers face a new task
- Seekers improve, then hiders face a new task
- Non-stationarity becomes a source of learning signal

$$\mathcal{M}_{k+1}^{hide} = \mathcal{M}^{hide}(\pi_{seek,k})$$
$$\mathcal{M}_{k+1}^{seek} = \mathcal{M}^{seek}(\pi_{hide,k+1})$$

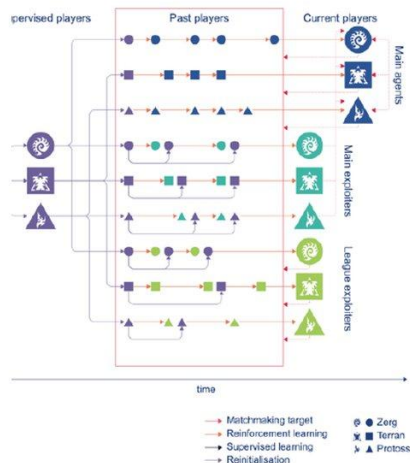
Hide-and-Seek paper: emergent strategy phases



# AlphaStar: League Training for Strategic MARL

- StarCraft II: partial observability, huge action space, long horizon
- League contains main agents, exploiters, and league exploiters
- Opponent sampling improves robustness across strategies
- Human replay pretraining bootstraps RL fine-tuning
- Evaluation focuses on robustness, not one current self-play matchup

AlphaStar paper: league agents and exploiters



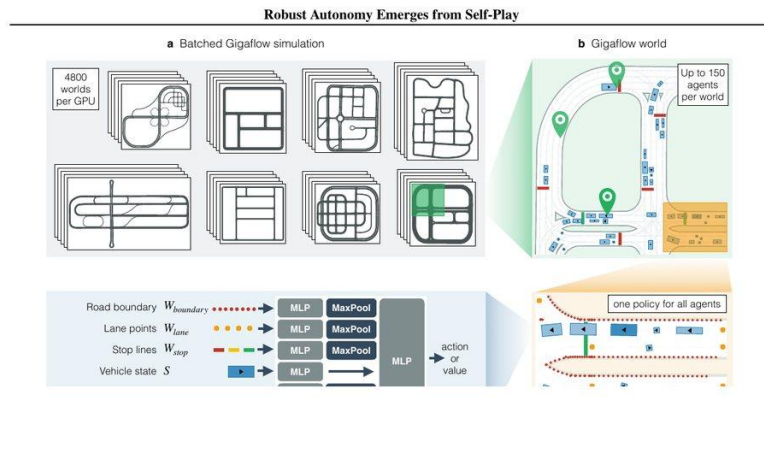
$$J_i(\pi_i) = \mathbb{E}_{\pi_j \sim \mathcal{L}} \left[ \sum_{t=0}^T \gamma^t r_i(s_t, a_t^i, a_t^j) \right]$$

$$\pi_j \in \{\text{main agents, exploiters, league exploiters}\}$$

# GigaFlow-Style Scaling

- Simulator throughput can dominate practical MARL
- More diverse games make stronger self-play possible
- Infrastructure needs workers, learner, policy pool, and evaluator
- Evaluation against populations is part of the system

GigaFlow paper: batched worlds and one policy for agents



$$\text{useful experience throughput} \approx N_{\text{env}} \cdot \text{steps/sec} \cdot q_{\text{diversity}}$$
$$\Delta J \propto \text{useful experience throughput}$$